Saurabh Kumar

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Bachelor of Technology
Computer Science and Engineering
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., CSE	National Institute of Technology, Patna	8.35	2028
Senior Secondary	Sachchidananda Sinha College,Aurangabad/BSEB	85.4%	2024
Secondary	Mahesh Academy,Aurangabad/CBSE	92.6%	2022

Projects

- **InterviewPilot AI — Full-Stack AI Platform** *June 2026 – Present*
  Tools: Next.js 16, TypeScript, MongoDB, Gemini AI, Upstash Redis, Razorpay, Cloudinary, JWT
  - Developed a full-stack platform utilizing **Next.js 16 App Router**, **TypeScript**, and **MongoDB** to deliver AI-driven resume evaluations, ATS scoring, and adaptive mock interviews.
  - Engineered an atomic credit wallet leveraging **MongoDB multi-document transactions** to guarantee data consistency across **Razorpay** payment fulfillments and AI operation debits.
  - Enforced server-side rate-limiting and progressive lockout logic using **Upstash Redis**, safeguarding authentication endpoints and AI routes against brute-force attacks.
  - Integrated **Google Gemini AI** with strict **Zod schema validation** and exponential-backoff retries, processing parsed PDF text into structured, deterministic ATS feedback reports.
- **MiniRedis: Asynchronous In-Memory Key-Value Store** *May 2026 – Present*
  Tools: C++17, Linux, Epoll, POSIX Sockets, RESP Protocol, Docker, CMake, Git
  - Developed an in-memory key-value database in **C++17** supporting **27 Redis commands** with full **RESP protocol** compliance.
  - Engineered an asynchronous **Reactor Event Loop** using Linux **epoll** and non-blocking sockets to handle concurrent clients efficiently.
  - Implemented hybrid **key eviction** with passive checks and **timerfd** sweeps, along with automated **RDB snapshot** persistence for data recovery.
  - Containerized using **Docker** and deployed on **Render** with a live web dashboard; ensured stability via **RAII** and graceful **SIGINT** shutdown.
- **DSA Saga – Web-based DSA Practice & Competing Platform** *[Team of 4]* *April 2026 – Present*
  Tools: Next.js, TypeScript, Docker, Tailwind CSS, MongoDB, Redis
  - **Interactive Practice Page & Code Editor:** Engineered the complete practice workspace with a custom multi-language code editor, live syntax highlighting, a Trie-based auto-completion engine, auto-save state caching, and live execution panels.
  - **Docker Code Execution Sandbox:** Architected an OOP-based containerized execution engine supporting 5 languages (C++, C, Python, Java, JS) with strict time/memory limit enforcement and automated public/hidden testcase evaluation.
  - **Admin Tooling & Testcase Generator:** Built the admin testcase management suite connected to MongoDB, featuring automated Docker-based testcase generation and side-by-side reference solution comparison.

Technical Skills and Interests

Languages: C, C++, Java, Python, SQL
Developer Tools: HTML, CSS, JavaScript, NodeJS, MongoDB,Docker, Git, GitHub, LaTeX
Frameworks: React, Next.js, Tailwind CSS
Areas of Interest: Competitive Programming, AI/ML, Web Development
Soft Skills: Teamwork, Time Management, English Speaking

Achievements

- **Competitive Programming**: Solved **1300**+ DSA problems across LeetCode (max rating **1767**), CodeForces (**1223**), and CodeChef (**1536**).